

Answer Key: Advanced Mastery Worksheet

Subject: Physics — Mechanical Oscillations

Exercise 1: Dynamic Equation & Kinematic Deduction

1.1 Differential Equation via Energy: The system is frictionless, so $ME = \text{constant}$.

$$ME = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = C \implies \frac{d(ME)}{dt} = 0.$$

$$\text{Differentiating: } \frac{1}{2}m(2vv') + \frac{1}{2}k(2xx') = 0 \implies mvv' + kxx' = 0.$$

$$\text{Since } v \neq 0 \text{ always, we divide by } v \text{ to get: } m\mathbf{x}'' + k\mathbf{x} = \mathbf{0} \implies \mathbf{x}'' + \frac{k}{m}\mathbf{x} = \mathbf{0}.$$

1.2 Verification of Solution: Given $x = X_m \cos(\omega_0 t + \varphi)$.

$$x' = v = -\omega_0 X_m \sin(\omega_0 t + \varphi).$$

$$x'' = a = -\omega_0^2 X_m \cos(\omega_0 t + \varphi) \implies x'' = -\omega_0^2 x.$$

$$\text{Substitute into the DE: } (-\omega_0^2 x) + \frac{k}{m}x = 0 \implies x(-\omega_0^2 + \frac{k}{m}) = 0.$$

$$\text{For this to hold for all } x, \omega_0^2 = \frac{k}{m} \implies \omega_0 = \sqrt{\frac{k}{m}}.$$

1.3 Kinematic Calculation:

$$- \omega_0 = \sqrt{\frac{40}{0.1}} = \sqrt{400} = \mathbf{20 \text{ rad/s.}}$$

$$- X_m = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2} = \sqrt{(0.02)^2 + \left(\frac{-0.4}{20}\right)^2} = \sqrt{0.0004 + 0.0004} = \sqrt{8 \times 10^{-4}} = \mathbf{2.83 \text{ cm (0.0283 m).}}$$

$$- \text{Phase } \varphi: \cos \varphi = \frac{x_0}{X_m} = \frac{0.02}{\sqrt{0.0008}} = \frac{1}{\sqrt{2}} \implies \varphi = \pm \frac{\pi}{4} \text{ rad.}$$

Since $v_0 < 0$, $\sin \varphi > 0$, we reject the negative angle. Thus, $\varphi = +\frac{\pi}{4} \text{ rad.}$

Exercise 2: Multi-Method Analysis from Energy Graphs

2.1 Curve Identification: **Curve A** is Elastic Potential Energy (EPE). It is 0 at equilibrium ($x = 0$) and maximum at the extremes ($x = \pm X_m$).

Curve B is Kinetic Energy (KE). It is maximum at $x = 0$ (where speed is highest) and 0 at the extremes (where the mass momentarily stops).

2.2 Graphical Extraction: The horizontal dashed line represents the total conserved energy. $ME = \mathbf{80 \text{ mJ (0.08 J)}}$.

The amplitude X_m is the abscissa where $PE = ME$ and $KE = 0$. From the graph, $X_m = \mathbf{4.0 \text{ cm}}$.

2.3 Energy Intersection: If $v = \frac{V_{max}}{\sqrt{2}}$, then $KE = \frac{1}{2}m\left(\frac{V_{max}}{\sqrt{2}}\right)^2 = \frac{1}{2}m\frac{V_{max}^2}{2} = \frac{1}{2}\left(\frac{1}{2}mV_{max}^2\right) = \frac{1}{2}ME$.

Since $ME = KE + PE \implies PE = ME - \frac{1}{2}ME = \frac{1}{2}ME$. Thus, **KE = PE** (intersection point).

$$PE = \frac{1}{2}ME \implies \frac{1}{2}kx^2 = \frac{1}{2}\left(\frac{1}{2}kX_m^2\right) \implies x^2 = \frac{X_m^2}{2} \implies x = \pm \frac{X_m}{\sqrt{2}}.$$

$$\text{For } X_m = 4.0 \text{ cm, } x_{int} = \pm \frac{4}{\sqrt{2}} = \pm \mathbf{2.83 \text{ cm.}}$$

Exercise 3: The Pseudo-Periodic Regime

$$\mathbf{3.1 Newton's Second Law: } \sum \vec{F} = m\vec{a} \implies \vec{T} + \vec{f} + \vec{N} + m\vec{g} = m\vec{a}.$$

$$\text{Projecting on the horizontal axis: } -kx - hv = ma \implies mx'' + hx' + kx = 0 \implies \mathbf{x'' + \frac{h}{m}x' + \frac{k}{m}x = 0}.$$

$$\mathbf{3.2 Power of Friction: } \frac{d(ME)}{dt} = \frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2\right) = mvv' + kxx' = v(mx'' + kx).$$

$$\text{From 3.1, } mx'' + kx = -hv. \text{ Substituting this yields } \frac{d(ME)}{dt} = v(-hv) = \mathbf{-hv^2}.$$

The negative sign signifies that mechanical energy is being continuously **dissipated** (lost to the environment as thermal energy due to friction).

Exercise 4: Linearization of the Phase Space

4.1 Derivation: $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kX_m^2$. Divide by $\frac{1}{2}m$: $v^2 + \frac{k}{m}x^2 = \frac{k}{m}X_m^2$.
Since $\frac{k}{m} = \omega_0^2$, we get $v^2 + \omega_0^2 x^2 = \omega_0^2 X_m^2 \implies \mathbf{v^2 = -\omega_0^2 x^2 + \omega_0^2 X_m^2}$.
Slope $\mathbf{m = -\omega_0^2}$; Y-intercept $\mathbf{c = \omega_0^2 X_m^2 = V_{max}^2}$.

4.2 Calculation:

- **a) Slope:** $m = \frac{0-0.25}{25 \times 10^{-4} - 0} = \frac{-0.25}{0.0025} = \mathbf{-100 \text{ s}^{-2}}$.
- **b) Angular Frequency:** $m = -\omega_0^2 \implies -100 = -\omega_0^2 \implies \omega_0 = \sqrt{100} = \mathbf{10 \text{ rad/s}}$.
- **c) Amplitude:** $c = V_{max}^2 \implies \omega_0^2 X_m^2 = 0.25 \implies 100 X_m^2 = 0.25 \implies X_m^2 = 0.0025 \text{ m}^2$.
 $X_m = \sqrt{0.0025} = 0.05 \text{ m} = \mathbf{5.0 \text{ cm}}$.